

Replication Plan

Effect of Single-Atom Platinum Doping and Facet Dependence on the Electronic Structure and Light Absorption of $\text{La}_2\text{Ti}_2\text{O}_7$: A DFT Study

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Paper Overview

This paper employs density functional theory (DFT) to investigate the effect of single-atom Pt doping on the electronic structure and optical absorption of monoclinic $\text{La}_2\text{Ti}_2\text{O}_7$ (LTO), a photocatalyst for water splitting. The study examines multiple surface facets ((010), (001), (100)), computes surface energies, analyzes band structures and density of states (DOS) upon Pt substitution at Ti sites, performs Bader charge analysis, and calculates optical absorption spectra using the DFT+U or hybrid functional approach.

Detailed Workflow

Phase 1: Structure Preparation

1.1 Obtain the $\text{La}_2\text{Ti}_2\text{O}_7$ crystal structure.

- From Materials Project or ICSD (monoclinic, space group $P2_1$).
- Optimize the bulk unit cell with DFT.

1.2 Construct surface slab models.

- Cleave (010), (001), (100) surfaces.
- Add vacuum layer ($\geq 15 \text{ \AA}$).
- Ensure stoichiometric and symmetric slabs.

1.3 Introduce Pt dopant.

- Substitute one Ti atom with Pt at various surface and subsurface sites.
- Create multiple doping configurations for each facet.

Phase 2: DFT Calculations

2.1 Geometry optimization.

- Relax all structures using PBE functional with DFT+U (or HSE06 for electronic properties).
- Convergence: forces $< 0.01 \text{ eV/\AA}$, energy $< 10^{-5} \text{ eV}$.
- VASP settings: PAW pseudopotentials, energy cutoff ($\sim 520 \text{ eV}$), k -point mesh (Gamma-centered).

2.2 Surface energy calculations.

$$\gamma = \frac{E_{\text{slab}} - n \cdot E_{\text{bulk}}}{2A}$$

2.3 Dopant formation energy.

$$E_f = E_{\text{doped}} - E_{\text{pristine}} - \mu_{\text{Pt}} + \mu_{\text{Ti}}$$

Phase 3: Electronic Structure Analysis

3.1 Band structure calculations along high-symmetry paths.

3.2 Projected density of states (PDOS) — decompose by atom and orbital.

3.3 Bader charge analysis to quantify charge transfer upon doping.

3.4 Charge density difference plots ($\Delta\rho$) for Pt–O bonding visualization.

Phase 4: Optical Properties

4.1 Compute dielectric function $\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega)$.

4.2 Derive absorption coefficient $\alpha(\omega)$ from the dielectric function.

4.3 Compare pristine vs. Pt-doped absorption spectra; quantify visible-light absorption enhancement.

Phase 5: Validation

5.1 Reproduce band gap values for pristine and doped systems.

5.2 Reproduce surface energy ranking across facets.

5.3 Reproduce DOS and absorption spectra plots.

5.4 Compare Bader charges to paper’s values.

Datasets

Dataset / Source	Type	Location / Notes
La ₂ Ti ₂ O ₇ crystal structure	CIF file	Materials Project: https://materialsproject.org/ (mp-3533 or similar)
PAW pseudopotentials	VASP POTCARs	Distributed with VASP license
Experimental band gap	Reference	Literature values for validation
Supplementary information	Tables, structures	Accompanying the paper

Models, Tools, and Software

Tool / Library	Version	Purpose / URL
VASP	5.4+ or 6.x	DFT calculations (geometry optimization, band structure, optics) https://www.vasp.at/
Quantum ESPRESSO	7.0+	Open-source DFT alternative https://www.quantum-espresso.org/
VESTA	≥ 3.5	Crystal structure visualization https://jp-minerals.org/vesta/
pymatgen	≥ 2022.0	Structure manipulation, VASP I/O https://pymatgen.org/
ASE	≥ 3.22	Atomic simulation environment

Tool / Library	Version	Purpose / URL
Bader analysis code	latest	Henkelman group Bader charge analysis http://theory.cm.utexas.edu/henkelman/code/bader/
VASPKIT	latest	Post-processing VASP outputs https://vaspkit.com/
Matplotlib	≥ 3.4	Plotting band structures, DOS, spectra
sumo	latest	Publication-quality band structure / DOS plots https://github.com/SMTG-Bham/sumo

Hardware Requirements

- **HPC cluster:** DFT surface slab calculations require ~ 16 – 64 cores, ≥ 64 GB RAM.
- **HSE06 calculations:** significantly more expensive; $\sim 100+$ cores, days per system.
- **Storage:** ~ 50 GB per slab calculation.